MapPoPE: what PoPE's polar score adds to a path-integrated phase (expansion note, 2026-10-20)

Labels: **ESTABLISHED** = registered result (results file named); **ANALYSIS** = new post hoc CPU analysis run for this note, eval-only on stored checkpoints (script and output named); **THEORY** = derivation, with its assumptions and a check against data; **SPECULATION** = untested.

Summary

1. **THEORY.** Both rotary scores have the form `s_ij = sum_c A_c cos(dphi_c + beta_c)`. RoPE and MapWM let content set the amplitude A and the phase beta. PoPE lets content set only A (softplus) and fixes beta = delta_c, a content-free bias. In our runs delta_c is effectively 0. So for every content pair, every PoPE head's kernel peaks at zero phase difference: s_ij <= sum_c A_c, with equality iff dphi_c = 0 mod 2 pi on every weighted channel.
2. With a **path phase**, "zero phase difference" means the same place, which is exactly the revisit target. PoPE's score removes MapWM's content phase, and MapWM's content phase misaims its kernel at the target. ANALYSIS: the kernel's alignment on the target key (C, defined in the theory section) is 0.70 for MapWM and 0.995 for MapPoPE (p 0.0002, 8/8). Removing the content phase alone would lift MapWM by +0.20. MapPoPE also spends 4x less attention on action keys.
3. With an **index phase**, "zero phase difference" means the query's own token. The 1-layer torus solution needs kernel peaks at fixed past offsets (4j-1). RoPE builds those peaks with its content phase; PoPE cannot. ANALYSIS: delta sits in the clamp's dead zone (98-100% of entries), K(3) > K(1) holds in 0% of PoPE content pairs, and 88% of the PoPE - RoPE gap (-0.110 of -0.126) is on multi-step reversals. This sign flip gives the +0.154 interaction, and about 80% of the interaction is the index side.
4. The score acts only on the readout term (C - C0). It never acts on the code term C0, which is how well the phase alone puts the same cell at zero. At T = 1024 and rank 2 (ANALYSIS), the score repairs short-gap revisits (C +0.34, p 0.0002; C0 unchanged). On wrap-only revisits C0 is 0.41-0.46 under both scores, and only rank 4 raises it. So no score rule can rescue rank 2 there.
5. Paper text: a short paragraph (Sec. 6 or App. L) and a 1-page appendix are proposed below. The single new GPU experiment is a delta-freedom 2x2 (index/path x delta pinned/free), about 2.5-3 GPU-hours. It predicts that the interaction reverses.

Established facts (registered unless marked)

fact	numbers	file
2x2 at T = 128, r2	RoPE 0.805, PoPE 0.679, MapWM 0.971, MapPoPE 0.999; interaction **+0.154** (MDE 0.076, 8/8). Encoding main effect -0.049. At r4 the interaction is +0.125, with both path arms at 1.000	`PAPER2X2_RESULTS.md`
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fact	numbers	file
Beyond training length the encoding helps the index arm	PoPE 0.639 / 0.607 vs RoPE 0.470 / 0.374 at T = 512 / 1024 (+0.169 / +0.233); r2 encoding main effect +0.114 / +0.189. At r4 the interaction is -0.172 / -0.221 (OOD; robustness, rule 10)	
T = 128 floors	blank 0.507, n-gram 0.598, **retrace 0.934** (wrap-only share 0). Both index arms sit below retrace	`docs/audits/2026-10-16/paper2x2_T128_floors_out.txt`
The gain is the score rule, not the frequency count	SCORE +0.0243 (p 0.012, 16/16 vs 10/16 SOLVED); COUNT +0.0002 (at ceiling, uninformative). CORRECTED: plausibly convergence speed; it shrank to +0.009 on fresh seeds	`MAPPOPE_PAIR_RESULTS.md`, `GAIN_GRAIN_RESULTS.md`
A peak-at-zero gain kernel is enough on a task with no headroom	GainScalar (one gain per token, delta 0) 20/20, the same as MapPoPE (descriptive: loss < 0.05 at epoch 23 vs 74 vs MapWM 132; confounded)	`GAIN_GRAIN_RESULTS.md`
No rank-2 rescue at T = 1024	3/12 vs 3/12 SOLVED. Secondaries: short gap +0.115 (p 0.036), gap >= 128 +0.043, wrap-only -0.015	`SCORE_RANK_RESULTS.md`
The score does not touch the what-to-where leak	GainRaw keeps MapWM's leak. ONE defect shown (CORRECTED)	`GAIN_PHASE_RESULTS.md`
Prior probes (post hoc)	MapWM r2 observation-key fields peak at d = 0 for 7% of content pairs (mean shift 13.7 cells); MapPoPE for 100%. Same-cell coherence C vs C0 at T = 128	`docs/audits/2026-10-05/remap_probe_out.txt`, `neuro_samecell_phase_out.txt`
Language / other	Bach MapPoPE - MapWM -0.0165 NLL (5/5); PoPE - RoPE -0.032. Code and Dyck MapPoPE rows are UNMEASURED or depth-OOD. PoPE faithful (1.7e-6); 80.7% of `pope_delta` in its clamp dead zone on the language runs	`JSB_RESULTS.md`, `.claude-memory/project_mappope_asymmetry.md`, `ABLATE_RESULTS.md`

Theory

THEORY 1 (the polar decomposition of a rotary score). Write channel c's query and key as complex numbers. RoPE / MapWM score: `s_ij = sum_c Re(q_c conj(k_c) e^{i dphi_c})` with `dphi_c = phi_c(i) - phi_c(j)`, so A_c = |q_c||k_c| and beta_c = arg q_c - arg k_c, both content-dependent. PoPE / MapPoPE (code: `model_pope.py`, `cos(dphi - delta)`): `A_c = softplus(q_c) softplus(k_c)`, beta_c = -delta_c, content-free. The phase dphi is the WHERE half: path integration changes where it comes from (`omega * cumsum(W_out W_in x)` instead of `t * theta`). The score map is the HOW half: whether content may enter the phase. The two halves are orthogonal design choices. This was already noted in `.claude-memory/reference_language_and_pope.md`; what follows makes it predictive.

Proposition 1 (peak pinning). If beta_c = 0 for all c, then for any content pair `s_ij <= sum_c A_c`, with equality iff dphi_c = 0 mod 2 pi on every channel with A_c > 0. As a function of displacement, the kernel is a non-negative combination of cosines, which is

positive-definite (Bochner). Content can reweight the frequencies, which changes the kernel's width and shape, but it cannot move or invert the peak. *Assumption:* delta = 0. **Check:** delta is in the clamp's dead zone (>= 0, zero gradient) on 98-100% of entries in every PoPE-Flat run, with mean |delta| = 0.00 (ANALYSIS A1). In MapPoPE C = C0 to 3 decimals (A1, A2), so delta does not shift the peak there either.

Corollary 1a (path phase). If the code is faithful (equal phase iff equal cell, within the head's resolved range), the pinned peak is "attend to where I am", which is the revisit target. A RoPE-form score must instead learn beta_c = 0 for every (action query, observation key) pair. A residual beta_c displaces the field in space by beta_c / omega_c, and the displacement is largest on the coarse channels. **Check (A1, T = 128, r2, n = 8):** target coherence is C = 0.703 for MapWM vs 0.995 for MapPoPE (p 0.0002, 8/8 seeds above). MapWM's content phase costs C0 - C = +0.20 (7/8 seeds). Same-cell attention mass is 0.747 vs 0.925 (p 0.0008).

Corollary 1b (key suppression must be by magnitude). Under a pinned peak, an action token at the same cell has the same phase as the observation there, so the score can only down-weight it through softplus(k). **Check:** mean softplus(k) is 0.61-1.56 for action keys vs 0.91-1.80 for observation keys. Attention on action keys is 0.041 for MapPoPE vs 0.179 for MapWM (p 0.0002). MapWM's amplitude-weighted cos(beta) is nearly the same for both key types (+0.76 vs +0.75), so it barely separates them in phase.

Corollary 1c (index phase). With an index phase, dphi = 0 is the query's own token. PoPE's frequencies are theta_c = 10000^(-c/64) <= 1 rad/token. Each cos(theta_c D) decreases on D in [0, pi], so with beta = 0 **every** content pair's kernel strictly decreases over offsets 0..3. A pinned-peak index head cannot prefer the observation 3 tokens back to the one 1 token back. Every fixed nonzero target offset must come from beta, which RoPE supplies through content and PoPE through delta (stuck at 0). The 1-layer torus task needs exactly such offsets: during a run that reverses the previous run, step j's target sits at token offset 4j-1 (the retrace rule, 0.934 at T = 128). **Check (A1):** - K(3) > K(1) holds in 0% of PoPE-Flat (head, action, observation) pairs on every seed, and in 53-100% for RoPE. - The argmax over D in 1..16 of the content-pair kernel is at D = 1 for 1068/1088 PoPE pairs. For RoPE it is at D = 3 (414), 7 (277), 10-11 (147) and 2 (168), a comb over the retrace offsets. - PoPE puts 0.307 of its attention on the newest observation (never the target; every step moves), vs 0.002 for RoPE (p 0.0002). - The accuracy gap decomposes over retrace strata as R1 -0.006, R2 -0.005, R3+ **-0.110**, non-retrace -0.004, out of -0.125. R3+ accuracy is 0.566 vs 0.779 (p 0.0002, 0/8 seeds overlap). The cost is entirely where the target is far back at a fixed offset; R1/R2, where the target is among the few most recent keys, are nearly intact.

Reading of the +0.154 interaction (THEORY + ANALYSIS). (MapPoPE - MapWM) - (PoPE - RoPE) = (+0.028) - (-0.126). The same constraint (peak pinned at zero phase difference) is aligned with the target under a path phase and misaligned under an index phase. About 80% of the interaction is the index-side loss; at r4 the path side is at ceiling and the interaction is +0.125, all index side. A pinned recency kernel is also length-agnostic. That fits PoPE's better extrapolation as an index code (0.607 vs RoPE 0.374 at T = 1024) and the PoPE paper's own remark that zero-init delta is "important for length generalization". This is a consistency check, not a test.

THEORY 2 (what a score rule can and cannot fix). Any score of this family sees position only through dphi. Write the target coherence as `C = C0 + (C - C0)`: - C0 is the code term: the A-weighted cos(dphi) on the true target, i.e. whether the phase alone puts the same cell at 0. - C - C0 is the readout term, which is beta's contribution.

A pinned-peak score sets the readout term to 0 by construction. It can also reweight channels (A >= 0), so it can **select** channels whose code is consistent but cannot **create** them. Wrap-only revisits need the code to be periodic on the torus, so they live in C0. **Check (A2, T = 1024, SCORE_RANK checkpoints, r2 n = 12, r4 n = 8):** - Short-gap revisits: C rises from 0.617 (MapWM) to 0.960 (MapPoPE), p 0.0002. C0 does not move (0.900 vs 0.960, p 0.16). Accuracy is +0.114 (p 0.036, replicating the registered +0.115). The repair is entirely the readout term. - Gap >= 128: C rises from 0.524 to 0.840 (p 0.029), but accuracy (+0.042, p 0.63) and target mass (-0.035) do not. C0 stays at 0.81-0.84 vs 0.99 at r4. The code has drifted, so many wrong-cell keys score as well as the target. Aiming the kernel exactly is not enough when the code cannot discriminate. - Wrap-only: C0 is 0.457 (MapWM) vs 0.411 (MapPoPE), p 0.79, and accuracy is 0.641 vs 0.631. Rank 4 raises MapWM's C0 to 0.840 (p 0.03) and accuracy to 0.998.

The score rule cannot rescue rank 2 because the rank-2 failure is in C0 (the code's periodicity and drift), which is outside any score's reach. Note: the score does not touch the code at T = 1024, but at T = 128 MapPoPE's C0 is also higher (0.996 vs 0.905, p 0.0002). A pinned peak may make a good code easier to *find* on short walks, consistent with GAIN_GRAIN's faster convergence; the effect does not carry to T = 1024 (SPECULATION as to mechanism).

Scope and assumptions. All checks are 1-layer, 2-head models on the 64x64 torus. In a 1-layer model Q and K depend only on the current token, so content cannot encode a computed offset. Deeper models could carry position into the amplitudes; PoPE's Indirect Indexing result (`INDIRECT_RESULTS_200k.md`, 7/8) is multi-layer and outside this account. "delta = 0" is a property of the faithful zero-init plus clamp. With delta ~ U(-2 pi, 0) the peak is still content-independent but no longer at zero, which is the lever of the proposed experiment.

Analyses run (all post hoc, eval-only, CPU)

All are in `/home/prashr/mapformer/docs/paper/expansion/scripts/mappope/`. - `kernel_probe.py` - Rebuilds layer 0 by hand; every run's logits equal the model's own to 0.0e+00 (`verify`). - Uses the same evaluation stream as `eval_noise_refine` / `eval_rank_strata` (held-out map seed 10000, np seed 1234 + s). Per-seed accuracies reproduce `runs/paper2x2/eval.log` exactly, e.g. RoPE s0 0.7892 and PoPE-Flat s0 0.6244. - A1 `paper2x2` (T = 128, 100 walks per seed, seeds 0-7): `kernel_probe_paper2x2_out.txt`, `.json`, `.log`. - A2 `score_rank` (T = 1024, **40** walks per seed, a subsample of the registered 100): `kernel_probe_score_rank_out.txt`, `.json`, `.log`. - Readouts: accuracy by retrace stratum (R1/R2/R3+/none) and by SCORE_RANK stratum (gap < 128 / >= 128 / wrap-only); attention mass on same-cell observation keys, the retrace target, the offset-1 observation, action keys and other observations; target coherence C and C0 on the most recent same-cell observation key; attention by token offset; content-pair kernel checks (K(3) > K(1), argmax offset, delta dead-zone share, zero-displacement score for action vs observation keys). - `probe_stats.py` -> `probe_stats_out.txt`: unpaired permutation tests (`stats_core.perm2_p`) on per-seed readouts,

and the stratum decomposition of PoPE - RoPE. - Caveats: - The masses and C use the head with the larger mean same-cell mass. In MapPoPE r4, wrap-only C0 is 0.55 while accuracy is 0.995, so another head or the FFN may carry wrap revisits there, and C0 is not a sufficient statistic for success. - A2 uses 40 walks per seed. - All p-values are post hoc and unregistered. They describe these checkpoints; they are not verdicts.

Proposed paper text

Main text, one paragraph (end of Sec. 6 or App. L lead):

> *Where the angle comes from and how content uses it are separate choices.* Any rotary score can be written `sum_c A_c cos(dphi_c + beta_c)`: the phase difference dphi says where, and content sets an amplitude A and, in RoPE, a phase beta. Path integration changes dphi. PoPE's polar score (Gopalakrishnan et al. 2025) removes beta. Without beta, every head's kernel peaks at zero phase difference whatever the content (App. L). Under a path phase, zero phase difference is the same place, which is the revisit target. Under an index phase it is the current token. The two predictions have opposite signs, and so do the results. On the short-walk torus PoPE's score raises the path-integrated model (MapPoPE 0.999 vs MapWM 0.971). There it aligns the attention kernel with the target (alignment 0.995 vs 0.70) and removes the cost of the content phase. It lowers the index model (PoPE 0.679 vs RoPE 0.805), whose solution needs kernel peaks at fixed past offsets. Nearly the whole loss (0.110 of 0.126) is on revisits where the target lies more than ten tokens back. A score rule acts only on how the kernel reads the phase, never on the phase itself. On long walks it repairs short-gap revisits (+0.115) and leaves wrap-around revisits, and the rank-2 deficit, untouched (3/12 vs 3/12 SOLVED): the rank-2 phase does not place a wrapped cell at zero.

Appendix L (replacing the current two lines), about one page: 1. The decomposition and Proposition 1, with the proof (one line) and its assumption (delta = 0; the dead-zone measurement). 2. Table L1, T = 128 2x2, one row per arm. Columns: accuracy; R3+ accuracy; target coherence C and C0; attention on the newest observation; attention on action keys; K(3) > K(1) share. Numbers are from `kernel_probe_paper2x2_out.txt` / `probe_stats_out.txt`. 3. The stratum decomposition of PoPE - RoPE (-0.006 / -0.005 / -0.110 / -0.004). 4. Table L2, T = 1024. C, C0 and accuracy by stratum for MapWM / PoPE score at r2 and r4. The readout term repairs short gaps; wrap-only lives in C0. 5. The registered facts (MAPPOPE_PAIR SCORE vs COUNT, SCORE_RANK NO RESCUE, GAIN_GRAIN no-headroom), each with its CORRECTED qualifier. 6. Label every probe number "post hoc, eval-only". Say plainly that the interaction is ~80% index side, and that the path-side gain is small (+0.024, shrinking to +0.009 on fresh seeds) and at T = 128 plausibly convergence speed.

What NOT to claim: that the polar score fixes rank or wrap; that MapPoPE beats MapWM at long walks (unmeasured at T = 1024, 3/12 vs 3/12); "PoPE hurts index models" in general (one layer, one task; PoPE helps index models on Bach and beyond training length here); "two defects, two fixes" (withdrawn).

Proposed experiments (GPU), ranked

1. **DELTA-FREEDOM 2x2 (the causal test of peak pinning).** - Design: phase {index, path} x delta {pinned: zero-init + clamp (current PoPE-Flat / MapPoPE-Flat); free: init U(-2 pi, 0), clamp removed}. Paper torus, T = 128, r2, 300 epochs cosine lr 1e-3, n = 8 fresh seeds, one batch (rule 12). Optional 5th arm: MapPoPE with delta frozen at U(-2 pi, 0), which removes the pin with no learning route back. - Primary readout (pre-register): the delta x phase interaction on T = 128 accuracy. - Predictions:
    - Free delta helps the index arm, concentrated in R3+ (a comb at offsets 4j-1 becomes reachable).
    - Free or frozen-random delta hurts or slows MapPoPE, since target coherence C falls below C0.
    - Secondary: free delta costs the index arm at T = 1024 (loss of the recency kernel).
    - Gates: delta dead-zone share and K(3) > K(1) share, eval-only with `kernel_probe.py`.
    - Cost: 4 arms x 8 seeds = 32 runs at about 38 min each (7.5 s/epoch at 4 jobs/GPU) is about 2.5-3 h on two GPUs, plus eval. The 5-arm version is about 3.5 h. A pilot on seeds outside the batch and a code-verification agent come first (rule 29).
    - Why first: it is the only test where the account predicts a sign reversal. It also turns the interaction from a description into a mechanism.
2. **Pinned-peak MapWM at T = 1024, r2.** Score `sum_c |q_c||k_c| cos(dphi_c)` (content phase removed, signed magnitude kept), plus MapWM and MapPoPE-Pair as twins, n = 12. Separates "beta removed" from "softplus magnitudes". Predicted: it matches MapPoPE on short gaps and does nothing on wrap-only. Cost about 36 runs at T = 1024 / 900 ep, which is substantial (SCORE_RANK-scale, about 1 day). Lower priority: GAIN_GRAIN already shows a pinned scalar gain suffices at T = 128.
3. **Eval-only (no GPU needed, can run now): polarise trained MapWM r4.** Replace beta by 0 at eval on the SCORE_RANK r4 checkpoints and read the accuracy change by stratum. `WHAT_WHERE_CHECKS` did the related "shared kernel x gain" rebuild at T = 128 (MapWM r2 0.971 -> 0.791). Expect small losses at r4 (C0 - C = 0.07) and large ones at r2. Diagnostic only, because the network was trained with beta.